

Assignment 3 Changes

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1 Changes Made

- Added `bufferpage` and `msgenrolled` to the `proc` structure in `kernel/proc.h`.
- Implemented `msgenroll()` by:
 - Allocating one page using `kalloc()`.
 - Clearing the page with `memset()`.
 - Mapping the page into the process page table.
 - Updating the process size.
 - Marking the process as enrolled.
- Implemented `msgsend()` by:
 - Checking that the message stayed within one page.
 - Looking through the process table to find the correct process.
 - Using `walkaddr()` to get the correct memory address.
 - Copying the message into the process buffer with `memmove()`.
- Implemented `msgread()` by:
 - Making sure the process was enrolled.
 - Checking that the requested data stayed inside the page.
 - Using `walkaddr()` to get the correct memory address.
 - Copying the data from the process buffer back to the user program.
- Initialized `bufferpage` and `msgenrolled` inside `allocproc()`.
- During testing, the code compiled but the program crashed with a kernel page fault. After debugging, I found that I was still using the original user pointers instead of the translated addresses from `walkaddr()`. Once I fixed that, the `share_msgs` program ran correctly and the parent and child processes were able to exchange messages.